hipot testers. EV motor test functions include inductance and resistance testing of windings, short-circuit testing between adjacent layers in the winding, and safety rating testing of the windings and armature. Voltage withstand test is required to meet the specifications of international electrical safety codes so that motor products can readily comply with their certification requirements.

For motor inductance tests, additional measurements of frequency, voltage and inductance are required. Inductance is measured with an AC signal, which is why an LCR meter is necessary. Layered short-circuit testing provides consistency during the motor manufacturing process by comparing waveforms. Wave comparison of a motor testing system include total area comparison, differential area comparison, flutter and corona.

Electromagnetic compatibility (EMC) testing

Electromagnetic shielding of cables constitutes an important concern in EV applications with considerations for developing particular standards for human exposure levels in radiated electromagnetic fields. It has been determined that the switching-frequency-related magnetic field radiation generated by power electronic converters using pulse width modulation (PWM) techniques can be of particular importance, overwhelming the fundamental frequency effects.

Simulation analysis of exposure levels due to variable frequency magnetic fields is done on anatomically-detailed human models in EV cabin environments. Testing performed using real-world simulations reduces cost and risk, and improves test coverage and overall design.

Increasing voltage of the traction battery or decreasing switching time of the power transistor to increase efficiency of an electric traction system results in an increase in the system's electromagnetic interference (EMI). To meet the challenges of new components, new measurement methods and new requirements considering EMI and immunity of HV components are continuously added to automotive standards.

An electric drive system consists of a high-voltage power source, an electric motor, a power converter, and shielded or unshielded high-power cables. These components set a path for electromagnetic emission. To avoid problems with EMI, extensive EMC components of vehicles are being developed. As electric cars need recharging after driving, it may cause problems when electromagnetic waves are over-generated while a battery is being recharged. So, electric cars need to be evaluated using two methods: driving mode with no charging on the power grid and rest in charging mode coupled to the power grid.

Some EV testing service and solution providers

Given below are some service and solution providers for EV testing.

- MET LAB provides EVSE testing, evaluation and certification to regional and international EV standards for safety and EMC. It determines what standards apply to a vehicle and develops a custom testing plan to help reach the intended markets efficiently and cost-effectively.
- Chroma provides automated test systems for EVs and HEVs, consisting of standard test platforms to test most power components in an EV's power system. Its automatic data recording and statistical report creation provide opportunities for product improvement. This combination of configurable hardware and software enables manufacturers to reduce cost and ensure consistency through all phases of the product lifecycle.
- Intertek offers lab capacity for EVs, HEVs and PHEVs, and automotive battery testing services on a global scale to certify charging stations and electrical components for worldwide distribution. From cells to large battery packs, the company offers performance, durability, safety and abuse testing to meet the latest industry, national and international standards and regulations.
- Scienlab supplies a hardware-in-the-loop test environment to emulate various cell types of a range of cell models. It accompanies the entire development and production process of an energy storage device.
- EVSELLC provides EV charger testers to ensure proper operation of equipment. It allows installers to check new EV chargers after installation, and verify a number of vital power and safety requirements, as these simulate the connection of an EV charger with an EV.
- Microtest provides motor tester series for performing static tests that measure electrical properties of a motor when it is not energised. It measures layered short-circuits, insulation voltage, inductance and DC resistance.
- NH Research's test equipment and software tools are used to validate designs and confirm manufacturing processes by using real-world simulations. Its battery emulation mode provides testing of chargers, DC/DC converters, inverters and vehicle-to-grid features.
- Seaward provides testers to ensure that all types of AC EVSEs are operating both safely and in accordance with IEC61851 at the point of installation and throughout their lifecycle. Testers can store diagnostic data supplied by the charge points, and data can be transferred wirelessly to an Android device running EVSE mobile app. Data can be sent to any remote location for in-depth fault diagnosis for traceability purposes. It can be used to measure output voltage of charge stations, maximum available charging current, earth loop impedance, RCD test, insulation resistance test, presence of correct mains supply and earth connection (single and three phases). It can also be used to simulate a number of fault conditions, disconnection time measurement, amplitude measurement, frequency measurement, duty cycle of PWM signal and so on.

To sum up

As the automotive industry advances in power conversion and battery technologies, it also improves EV testing performance and cost by developing flexible automated power conversion test platforms and regenerative battery test systems. Automated test solutions provide safe and real-time monitoring of test processes and reduces inspection time, significantly reducing production costs. EFY